

Mason Brooks

Senior Full Stack Developer

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SUMMARY

Senior Full Stack Engineer with 10 years of experience building, scaling, and optimizing enterprise-grade web applications using JavaScript, TypeScript, and Python. Strong expertise across modern frontend and backend frameworks including React, Angular, Vue.js, Node.js, Django, and FastAPI, with a solid foundation in API architecture, microservices, and system design. Proven hands-on experience with AWS and Azure, CI/CD pipelines, Docker, and cloud-native deployments. Highly effective in Agile/Scrum environments, collaborating with cross-functional teams to deliver secure, performant, and maintainable solutions. Recognized for solving complex technical problems and driving products from concept to production with measurable business impact.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2009 – May 2013

University of Texas at Austin, Austin, TX

WORK EXPERIENCE

Senior Full Stack Developer | Electric Power Engineers

May 2023 – Dec 2025 | Austin, TX

- Led development of a cloud-hosted full-stack platform, designing backend services with **Python (Django)** and **FastAPI microservices** and building interactive frontends using **React, JavaScript, TypeScript, Tailwind CSS, Bootstrap, and SCSS**.
- Architected and delivered **secure, scalable APIs** using **RESTful APIs, GraphQL**, and **gRPC**, supporting geospatial visualization of U.S. electricity usage, power station metadata, and transmission line analytics.
- Designed geospatial and data processing services integrating third-party grid and energy APIs, enabling real-time and aggregated usage calculations with dynamic, threshold-based map rendering.
- Built and optimized **PostgreSQL data models**, ensured efficient data access, and improved end-to-end performance through backend query optimization and frontend state management.
- Implemented enterprise authentication and authorization, including **SSO** with **Keycloak, OAuth 2.0 / JWT**, and **OpenFGA-based RBAC & ReBAC**, securing sensitive energy datasets and application features.
- Ensured secure and reliable service-to-service communication, implementing request validation, standardized error handling, and token-based authorization across distributed microservices.
- Containerized and deployed services using **Docker** and **Kubernetes**, implemented **CI/CD pipelines** aligned with the **SDLC**, and operated workloads across **Azure, AWS, and GCP**.
- Collaborated closely with QA, Data Science, and Product teams, demonstrating strong **communication skills** through clear technical documentation, cross-team alignment, and active participation in **Agile/Scrum** processes.

Senior Software Engineer | CX Score

Oct 2019 – Apr 2023 | Mountain View, CA

- Full-stack web application development of an analytics platform evaluating **Core Web Vitals, SEO, accessibility**, and performance metrics to identify optimization opportunities for user-submitted websites.
- **Backend architecture** and **API orchestration** using **Python** and **Django** to manage website ingestion pipelines, analysis workflows, scoring logic, and automated optimization recommendations.
- **RESTful API** development supporting performance analysis, historical data persistence, and frontend data visualization for analytics dashboards.
- **Backend data processing pipelines** analyzing **Core Web Vitals (LCP, CLS, INP, FCP, TTFB)** and transforming audit data into normalized performance scores and prioritized insights.
- Asynchronous background task engineering to support long-running website audits, improving system throughput and responsiveness.
- **Relational data modeling** to persist audit results, metric snapshots, and historical trend data for longitudinal analysis.
- Backend performance tuning through **database query optimization, caching strategies, and API response improvements** to support concurrent workloads.
- React-based frontend development rendering external websites securely and presenting **performance, accessibility, and SEO analytics** in a unified dashboard.
- Frontend optimization techniques including code splitting, lazy loading, memorization, and asset optimization to improve rendering efficiency.
- SEO and accessibility compliance integration aligned with **Google Lighthouse** and **WCAG 2.1** standards.
- **Agile** cross-functional collaboration supporting sprint planning, feature refinement, and iterative platform improvements.

Software Engineer | Diffco

Jun 2013 – Sep 2019 | San Jose, CA

- Scalable full-stack system architecture design supporting high-traffic booking workflows and complex internal operations across multi-location businesses.
- **Service-oriented backend** architecture using **Node.js, Express** and **NestJS**, enabling modular workflows for scheduling, payments, notifications, and participant lifecycle management.
- Dynamic pricing and availability engine development using **PostgreSQL/MySQL**, handling capacity rules, maintenance windows, and location-specific constraints.
- Mobile-first Frontend engineering with **React, Redux, Next, JavaScript (ES6+), TypeScript, Tailwind CSS, Bootstrap, HTML5**, and **SCSS**, optimizing performance and conversion rates.
- Secure payment system integration with **Stripe, PayPal, Apple Pay**, supporting refunds, reconciliation, and order lifecycle tracking.
- Event-driven communication systems using **Twilio** and **SendGrid** for transactional emails, SMS notifications, and customer engagement workflows.
- Internal **ERP** module development supporting room configuration, staff operations, live monitoring, and reporting systems.
- Analytics and tracking integration using **Google Analytics** and **Facebook Pixel** for **KPI** tracking and funnel analysis.
- Application and database **performance optimization** improving API response times and system reliability during peak traffic.
- Containerized deployment and infrastructure management using **Docker, Nginx, and Linux**.
- **Agile/Scrum** collaboration translating business requirements into scalable technical solutions.
- Code quality and maintainability enforcement through Git workflows, code reviews, and continuous refactoring.

PROJECTS

Confidential / High-Traffic Entertainment & Booking Platform - Quest Room

<https://questroom.com>

Designed a scalable, multi-location booking platform with dynamic scheduling, pricing, and secure payments. Built a modular Node.js backend and a high-performance, mobile-first React frontend. Implemented event-driven notifications, analytics, and operational tooling to support high-traffic demand.

SKILLS

Programming Languages

JavaScript (ES6+), TypeScript, Python, HTML5, CSS3

Frontend Technologies

React, Next.js, Angular, Vue.js, Redux, Zustand, React Query, Webpack, Vite, Babel, Tailwind CSS, SCSS, Responsive Design, Mobile-First Design, Cross-Browser Compatibility, Component-Driven Architecture, UI Performance Optimization

Backend & API Development

Django, FastAPI, Flask, Node.js, Express.js, NestJS, RESTful APIs, GraphQL, API Versioning, API Security, Microservices Architecture, Event-Driven Systems, Background Workers, Task Scheduling

Cloud, DevOps & Infrastructure

AWS (EC2, S3, RDS, Lambda, CloudWatch, IAM), Azure (App Services, Azure Functions, Azure DevOps), Docker, Kubernetes, Helm, CI/CD Pipelines (GitHub Actions, GitLab CI), Infrastructure as Code (Terraform), Linux, Nginx, Apache, Deployment Automation

Databases & Data Technologies

PostgreSQL, MySQL, Oracle, MongoDB, Redis, Elasticsearch, Prisma ORM, TypeORM, Sequelize, SQLAlchemy, Django ORM, Database Schema Design, Query Optimization, Indexing Strategies, Transaction Management, Data Migration

Performance & Scalability

Core Web Vitals, Frontend Performance Optimization, Backend Profiling, Load Optimization, Caching Strategies (Redis, CDN), Asynchronous Processing, Parallel Processing, Memory Optimization, CPU Optimization, High-Availability Systems, Fault-Tolerant Architecture

Security & Authentication

OAuth 2.0, JWT, Role-Based Access Control (RBAC), Relationship-Based Access Control (ReBAC), Secure Coding Practices, API Authentication, Authorization Strategies

Testing & Quality

Jest, Mocha, PyTest, Unit Testing, Integration Testing, API Testing, Test-Driven Development (TDD), Code Quality Standards

Version Control & Collaboration

Git, GitHub, GitLab, Bitbucket, Git Flow, Branching Strategies, Pull Requests, Code Reviews, Release Management, Semantic Versioning

Methodologies & Leadership

Agile Methodology, Scrum, Cross-Functional Collaboration, Technical Leadership, Mentorship, System Design, Problem Solving, Requirements Analysis